

respectively. Notice the effect of trading costs on portfolio performance. This cost dominates the effect of the timing risk of the strategy.

Important Note: The investor will not be able to achieve the expected portfolio return even via a very passive strategy such as VWAP because even a very passive VWAP strategy will incur permanent impact cost. The portfolio risk calculation is sub-additive whereas portfolio variance is an additive relationship. This results in trading costs primarily due to market impact cost having a much more dramatic effect on portfolio performance than corresponding timing risk of the strategy.

Which execution strategy should the trader use?

As stated above, some investors may wish to trade passively, such as with strategy γ , to minimize market impact cost and some may wish to trade aggressively, such as with strategy z , to minimize timing risk or possibly lock in a market mispricing or realized profit. Further, there are other investors who prefer a strategy somewhere in the middle, such as strategy x . But which strategy is most appropriate? Since each of these strategies lie on the efficient trading frontier can they all be considered a best execution strategy?

The answer is no. There is only a single best execution strategy. (This answer may surprise some readers.)

Our conclusion is described following the same investor utility maximization that was used to determine the preferred optimal portfolio A1 on the efficient trading frontier.

Combining the efficient investment and efficient trading frontiers onto one chart provides investors with the ability to determine the proper optimal strategy for a specified portfolio. To show this, first recall that investors determine their preferred optimal portfolio through maximizing their utility function. But now let us maximize investor utility for both frontiers. This is illustrated in [Figure 10.5](#).

The portfolio manager selected portfolio A1 as the preferred strategy because it was the portfolio that maximized the investor's utility function (shown as U_2 in [Figure 10.2](#)). Now let us apply the same technique used to determine portfolio A1 to determine the optimal trading strategy. First, utility U_4 passes through two strategies on the efficient trading frontier: aggressive strategy z and passive strategy γ . Since both of these strategies lie on the same indifference curve they provide the investor with equal utility. This may be surprising to many. This analysis shows that two seemingly opposite strategies (aggressive and passive) can have the same